

EAE 129 WQ 2025 Final Project:
Simulation and Analysis of Aircraft Response
characteristics

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1 Abstract

Performing the wind tunnel test of the prototype at sea level at Mach (M) 0.257, we analyzed the static stability of the fighter jet and found all aerodynamic coefficients to design a suitable controller. After obtaining the aerodynamic characteristics, we used Matlab to analyze aircraft dynamic stability and flying qualities. We derived stability derivatives from aircraft and stability data, set up a state-space model, simulated and plotted time responses to control inputs, identified aircraft dynamic modes, analyzed the dynamic response characteristics and flying quality, and performed reduced order analysis to approximate Phugoid and Short-period modes.

2 Introduction

A fundamental principle in aerodynamics and flight dynamics is that aircraft undergo extensive testing with prototype models to evaluate their performance before the first flight. These prototypes, commonly known as mock-ups, are designed to have aerodynamic characteristics similar to those of the final aircraft. In our wind tunnel experiment, our goal is to visualize the aerodynamic characteristics of a new fighter jet in numerical values and provide experimental data that control engineers can utilize to design a control system for the aircraft. The results and derivations presented in this report are intended to be reproducible for further analysis. We assume that the aircraft's motion consists of small deviations from its equilibrium flight condition. Additionally, we assume that the aircraft's motion can be analyzed by separating the governing equations into two groups:

- The **longitudinal equations** consist of the X -force, Z -force, and pitching moment equations.
- The lateral equations consist of the Y -force, rolling moment, and yawing moment equations.

Since this experiment focuses primarily on longitudinal analysis, we do not consider the Y -force in our calculations.

3 Results

3.1 State Space Form

The equation of rigid-body motion in state-space form is given by:

$$\dot{\vec{x}} = A\vec{x} + B\vec{u} \quad (1)$$

In the form of Equation 1, the longitudinal linearized equation of motion in state-space form is given by:

$$\begin{aligned}
\begin{bmatrix} \Delta \dot{u} \\ \Delta \dot{w} \\ \Delta \dot{q} \\ \Delta \dot{\theta} \end{bmatrix} &= \begin{bmatrix} X_u & X_w & 0 & -g \\ Z_u & Z_w & u_0 & 0 \\ M_u + M_{\dot{w}}Z_\delta & M_w + M_{\dot{w}}Z_w & M_q + M_{\dot{w}}Z_\delta & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta \theta \end{bmatrix} \\
&+ \begin{bmatrix} X_\delta & X_{\delta_T} \\ Z_\delta & Z_{\delta_T} \\ M_\delta + M_{\dot{w}}Z_\delta & M_{\delta_T} + M_{\dot{w}}Z_{\delta_T} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \Delta \delta \\ \Delta \delta_T \end{bmatrix}
\end{aligned} \tag{2}$$

In this report, we do not consider the elevator deflection angle. Therefore, $\Delta \delta_T$ term can be neglected and Equation 2 can be reduced to:

$$\begin{aligned}
\begin{bmatrix} \Delta \dot{u} \\ \Delta \dot{w} \\ \Delta \dot{q} \\ \Delta \dot{\theta} \end{bmatrix} &= \begin{bmatrix} X_u & X_w & 0 & -g \\ Z_u & Z_w & u_0 & 0 \\ M_u + M_{\dot{w}}Z_\delta & M_w + M_{\dot{w}}Z_w & M_q + M_{\dot{w}}Z_\delta & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta \theta \end{bmatrix} \\
&+ \begin{bmatrix} X_\delta \\ Z_\delta \\ M_\delta + M_{\dot{w}}Z_\delta \\ 0 \end{bmatrix} [\Delta \delta]
\end{aligned} \tag{3}$$

Note the notations used in Figure 7 are slightly different from those in Equation 3. For convenience, we treat δ and δ_e as the same variable. **Now determine matrix A and matrix B in numerical values** using Equation 3, formulas in Figure 7, the aerodynamic data of the fighter jet performing at sea level (5), and following assumptions and designs made for the report:

1. $X_{\delta_T} = 0$
2. $C_{L_u} = 0$
3. $C_{D_u} = 0$
4. $C_{m_u} = 0$
5. $c = \bar{c}$
6. $C_{L_0} = C_L$
7. $C_{D_0} = C_D$
8. $C_{Z_{\delta_e}} = -C_{L_{\alpha_t}} \tau \eta \frac{S_t}{S}$ where
 - $C_{L_{\alpha_t}}$, horizontal tail lift curve slope, is 3,
 - τ , the elevator effectiveness, is 0.5,
 - η , the dynamic pressure ratio, is 0.95, and
 - S_t , the area of the horizontal tail, is 50 ft^2 .

Using MATLAB, we obtained the following A and B matrix from our experiment data:

$$A = \begin{bmatrix} -0.0694 & 0.0376 & 0 & -32.1700 \\ -0.1940 & -0.4887 & 286.9148 & 0 \\ 0.0001 & -0.0068 & -0.3850 & 0 \\ 0 & 0 & 1.0000 & 0 \end{bmatrix} \quad (4)$$

$$B = \begin{bmatrix} 0 \\ 13.7579 \\ -4.5681 \\ 0 \end{bmatrix} \quad (5)$$

4 Discussion

4.1 Model Characteristics

1. Find out the eigenvalues of the A matrix.

Using MATLAB, we have obtained two pairs of complex conjugates.

$$\begin{aligned} \lambda_{1,2} &= -0.440 \pm 1.3952i \\ \lambda_{3,4} &= -0.0316 \pm 0.1391i \end{aligned}$$

Table 1: Eigenvalues

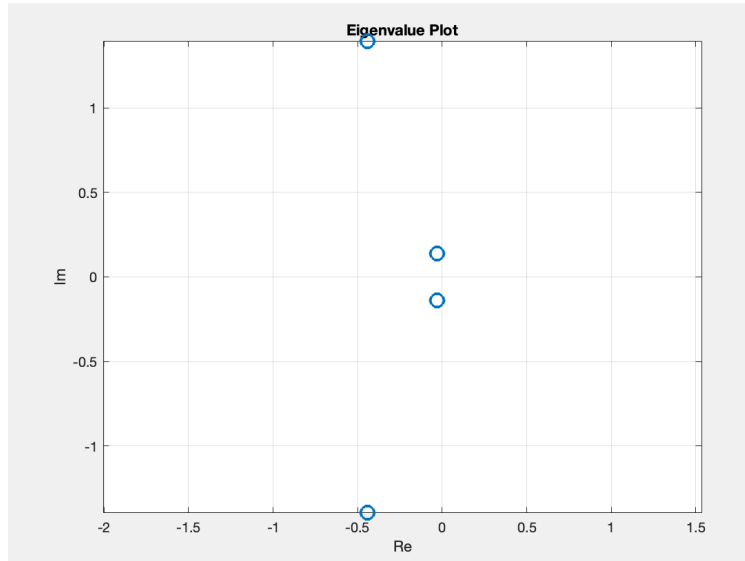


Figure 1: Characteristic roots/eigenvalues

2. Identify the eigenvalues corresponding to the two longitudinal modes. Determine each mode's damping ratio, neutral frequency, damped frequency, and time constant.

When disturbed from its equilibrium flight condition, an airplane's longitudinal motion is characterized by two oscillatory modes of motion. The first mode is lightly damped with a long period, known as the **Phugoid mode**. The second mode is heavily damped with a very short period, referred to as the **Short-Period mode**. All airplanes exhibit these two longitudinal dynamic modes, and the aircraft's response is a linear combination of both. The Phugoid conjugate pair typically exists near the origin O , while the short-period pair is located farther from the origin. Therefore, we can conclude that:

$$\begin{aligned}\text{Short Period Mode} &= -0.440 \pm 1.3952i \\ \text{Phugoid Period Mode} &= -0.0316 \pm 0.1391i\end{aligned}$$

Table 2: Longitudinal Modes

We use the same method to calculate aerodynamic responses for the two modes:

$$w_n = \sqrt{\text{Re}(\lambda)^2 + \text{Im}(\lambda)^2} \quad (6)$$

$$\zeta = -\frac{\text{Re}(\lambda)}{w_n} \quad (7)$$

$$w_d = \text{Im}(\lambda) = w_n \sqrt{1 - \zeta^2} \quad (8)$$

$$\tau = -\frac{1}{\text{Re}(\lambda)} \quad (9)$$

Using the list of Equations above and characteristic roots from Table 2, we have:

(a) Phugoid Period Mode

Model Characteristic	Result
Damping ratio : ζ_p	0.2216
Natural frequency : w_{np}	0.1426 <i>rad/s</i>
Damped frequency : w_{dp}	0.1391 <i>rad/s</i>
Time constant : τ_p	31.6492 <i>s</i>

Table 3: Phugoid Period Mode

(b) Short Period Mode

Model Characteristic	Result
Damping ratio : ζ_{sp}	0.3007
Natural frequency : w_{nsp}	1.4629 <i>rad/s</i>
Damped frequency : w_{dsp}	1.3952 <i>rad/s</i>
Time constant : τ_{sp}	2.2729 <i>s</i>

Table 4: Short Period Mode

4.2 Time Response

- Use MATLAB to plot the time responses of Δu , Δw , Δq , and $\Delta \theta$ for the following scenarios.

1. Zero input and non-zero initial condition $x_0 = [0, 0, 0, 0.1]^T$.

In other words, we examine the longitudinal motion of an airplane without control input; It is a free response (controls fixed).

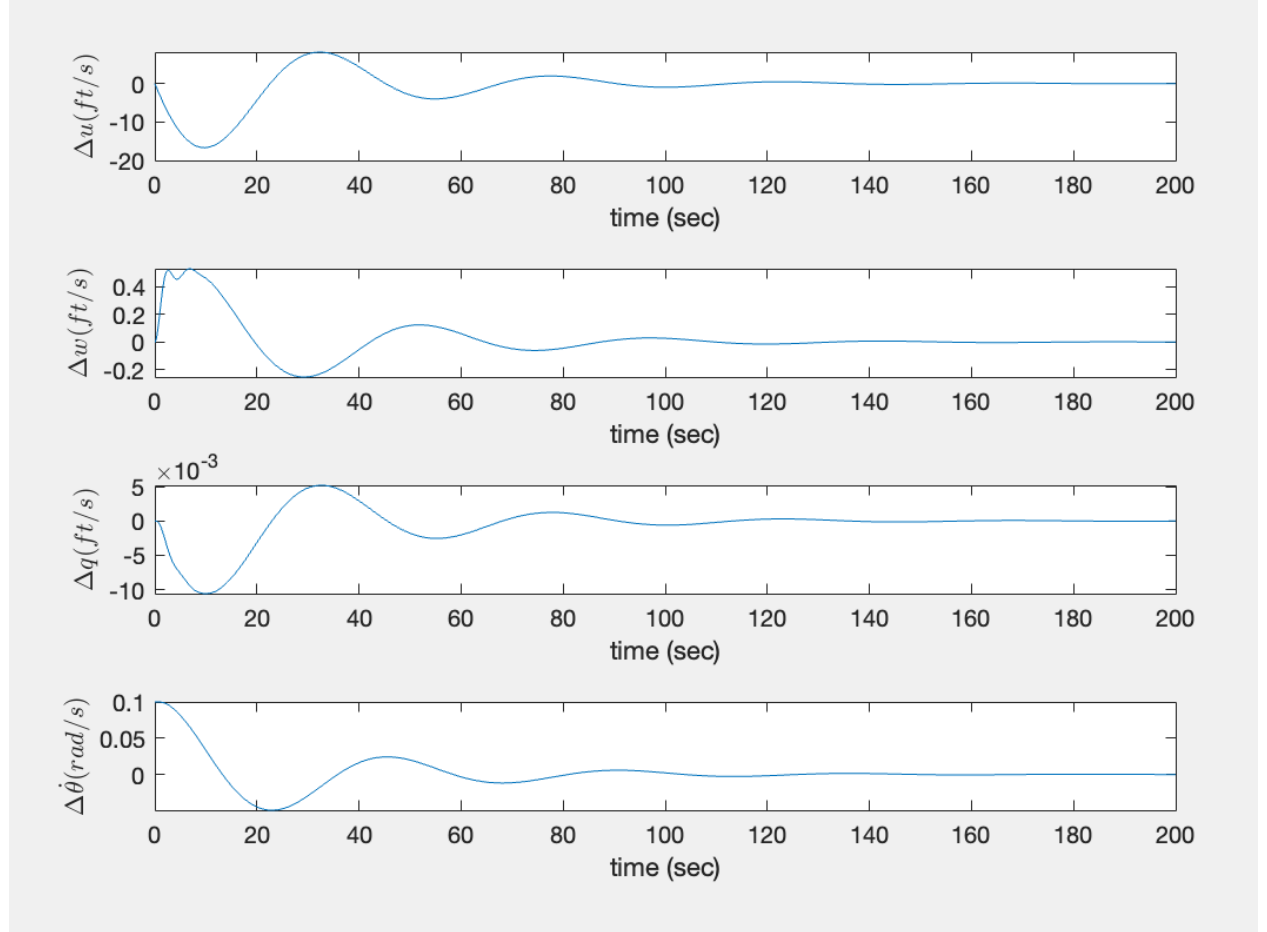


Figure 2: Free response with a non-zero initial condition

As expected, all the five control outputs show decaying trends to zero and they all gradually become steady in the free-response simulation. The MATLAB simulation demonstrates that the aircraft exhibits the expected longitudinal modes: a lightly damped Phugoid oscillation and a more heavily damped Short-Period response. The Phugoid mode's slow oscillation with low damping is clearly shown in the three directional velocities as they go roughly two cycles to become steady.

Future improvements could focus on increasing Phugoid damping ratio for more stability.

2. Zero initial condition and a unit step elevator input.

Since we only have one input, $\Delta\delta_e$, in our state-space model, we increase $\Delta\delta_e$ by 1 with a step response for any $t \geq 0$.

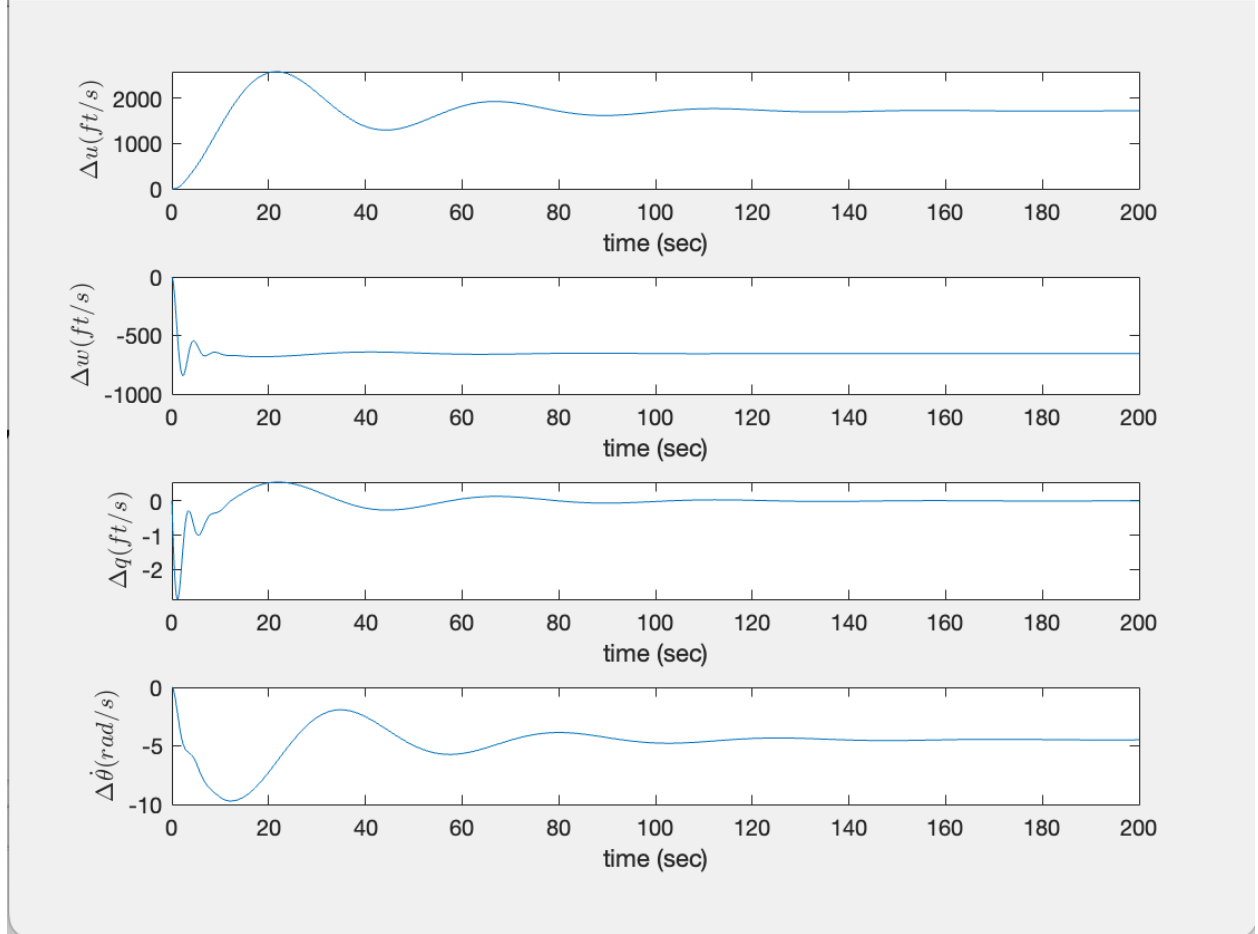


Figure 3: Step response with a zero initial condition

The MATLAB simulation exhibits a steady decaying response for a unit step input and confirms that the Short-Period mode plays a dominant role in pitch attitude correction as the Short-Period mode decays quickly. This is evident in $\Delta\dot{w}$ and $\Delta\dot{q}$, where oscillations damp out rapidly, indicating a well-damped response of short-period mode. In contrast, $\Delta\dot{u}$ and $\Delta\dot{\theta}$ exhibit slow decay with a longer period, suggesting low damping in the Phugoid mode. This slower oscillation affects the long-term response, potentially increasing pilot workload and influencing handling qualities. While the aircraft responds quickly to control inputs, prolonged Phugoid oscillations may require active damping mechanisms to improve stability.

Again, in order to improve the performance, we would need to increase phugoid damping.

4.3 Handling Qualities Investigation

- Determine the flying level given that the aircraft is of Class IV and flies in Category B.

According to Table 8, for the Phugoid mode, our damping ratio ζ is 0.2216, which is greater than 0. Based on Table 8, a Phugoid mode with $\zeta > 0.2$ corresponds to either Level 1 or Level 2 flying quality. For the Short-Period mode, in the Category B column of Table 8, our damping ratio $\zeta_{sp} = 0.3007$ exceeds the minimum threshold of 0.02 for Level 1. Since both modes satisfy at least Level 1 requirements, we conclude that the aircraft's flying quality level is Level 1.

Flying Quality Level : Level 1

- Level 1 : Flying qualities clearly adequate for the mission flight phase.
- Class IV : High-maneuverability airplanes, such as fighter/interceptor.
- Category B : Non-terminal flight phases that are normally accomplished using gradual maneuvers and without precision tracking, although accurate flight-path control may be required.

The flying qualities satisfy our expectation for our fighter jet.

4.4 Reduced Order Analysis

As shown in the free and step responses in Figures 2 and 3, two distinct trends emerge, allowing us to further approximate the equations of motion (EOM).

The state-space form of the two-degrees-of-freedom short-period approximation can be obtained by assuming $\Delta u = 0$ and dropping the X -force equation.

$$\begin{bmatrix} \Delta \dot{w} \\ \Delta \dot{q} \end{bmatrix} = \begin{bmatrix} Z_w & u_0 \\ M_w + M_{\dot{w}}Z_w & M_q + M_{\dot{w}}u_0 \end{bmatrix} \begin{bmatrix} \Delta w \\ \Delta q \end{bmatrix} \quad (10)$$

The long-period mode is characterized by changes in pitch attitude, altitude, and velocity at a nearly constant angle of attack. An approximation to the two-degrees-of-freedom long-period mode can be obtained by neglecting the pitching moment equation and assuming that the change in angle of attack is 0.

$$\Delta \alpha = \frac{\Delta w}{u_0} \rightarrow \Delta \alpha = 0 \rightarrow \Delta w = 0 \quad (11)$$

Therefore, the longitudinal state equations (3) reduce to:

$$\begin{bmatrix} \Delta \dot{u} \\ \Delta \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & -g \\ \frac{-Z_u}{u_0} & 0 \end{bmatrix} \begin{bmatrix} \Delta u \\ \Delta \theta \end{bmatrix} \quad (12)$$

1. Find out the eigenvalues of the A matrix for the short-period approximation 10, and the phugoid approximation 12.

Using MATLAB, we first obtain the A matrix for Phugoid and Short-period approximations:

$$A_{phugoid} = \begin{bmatrix} -0.0694 & -32.1700 \\ 0.0007 & 0 \end{bmatrix} \quad (13)$$

$$A_{short} = \begin{bmatrix} -0.4887 & 286.9148 \\ -0.0068 & -0.3850 \end{bmatrix} \quad (14)$$

Now, we obtain eigenvalues from the matrices above:

$$\begin{aligned} \text{Short-Period Approximation} &= -0.4369 \pm 1.3991i \\ \text{Phugoid Period Approximation} &= -0.0347 \pm 0.1433i \end{aligned}$$

Table 5: Approximated eigenvalues

2. Compare the natural frequencies and damping ratios from Subsection 4.1 to those obtained through approximations and evaluate the accuracy of the approximation.

Additionally, we use Equations 6, 8, and 7 to compute the aerodynamic responses from the approximated state-space system.

(a) Phugoid Period Mode Comparison

Model Characteristic	Exact method	Approximate method	Difference
Damping ratio : ζ_p	0.2216	0.2434	8.9565 %
Natural frequency : w_{n_p}	0.1426 <i>rad/s</i>	0.1475 <i>rad/s</i>	3.322 %
Damped frequency : w_{d_p}	0.1391 <i>rad/s</i>	0.1433 <i>rad/s</i>	2.931 %

Table 6: Phugoid Period Mode Comparison

(b) Short Period Mode Comparison

Model Characteristic	Exact method	Approximate method	Difference
Damping ratio : ζ_{sp}	0.3007	0.2986	0.7032 %
Natural frequency : $w_{n_{sp}}$	1.4629 <i>rad/s</i>	1.4657 <i>rad/s</i>	0.1910 %
Damped frequency : $w_{d_{sp}}$	1.3952 <i>rad/s</i>	1.3991 <i>rad/s</i>	0.2788 %

Table 7: Short Period Mode Comparison

The approximations used in the reduced-order analysis, particularly for the Phugoid mode, introduced discrepancies when compared to full-state modeling (3). The deviation in

natural frequency and damping ratio (with errors of 8.9565 % and 3.322 %, respectively) suggests that the long-period behavior of the aircraft may be more sensitive to simplifications than the short-period mode. This highlights the need for higher-fidelity models in precise control design, especially for highly maneuverable aircraft, such as our fighter jet.

The short-period mode is more accurate.

5 Conclusion

In this report, we analyzed the longitudinal dynamics of the aircraft, focusing on the Phugoid and Short-Period modes. The reduced-order analysis provided a useful approximation of the aircraft's dynamics, but discrepancies, particularly in the Phugoid mode damping ratio, indicate that such models may not fully capture the system's long-period behavior. The observed deviations in natural frequency and damping ratio highlight the need for higher-fidelity models in precise control design, especially for highly maneuverable aircraft such as fighter jets. For future improvements, the mass distribution could play a critical role in Phugoid damping, as adjusting the center of gravity and overall weight distribution can enhance stability and mitigate long-period oscillations. Also, adding small aerodynamic modifications to increase profile drag, C_D could increase the damping ratio. Future work could focus on optimizing these parameters alongside aerodynamic refinements to improve both accuracy and performance. Overall, the aerodynamic data obtained from the wind tunnel experiment effectively captured the aircraft's characteristics, providing valuable insights into its dynamic behavior.

6 Appendix

6.1 Longitudinal Derivatives

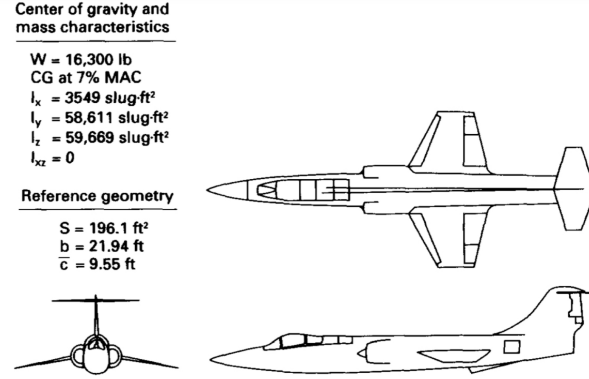


Figure 4: Characteristics of the fighter aircraft

6.2 Aerodynamic Data

Longitudinal	C_L	C_D	$C_{L\alpha}$	$C_{D\alpha}$	$C_{m\alpha}$	$C_{L\dot{\alpha}}$	$C_{m\dot{\alpha}}$	C_{Lq}	C_{mq}	C_{LM}	C_{DM}	C_{mM}	$C_{L\dot{\delta}_e}$	$C_{m\dot{\delta}_e}$
M = 0.257														
Sea level	0.735	0.263	3.44	0.45	-0.64	0.0	-1.6	0.0	-5.8	0.0	0.0	0.0	0.68	-1.46
M = 1.8														
55,000 ft	0.2	0.055	2.0	0.38	-1.30	0.0	-2.0	0.0	-4.8	-0.2	0.0	-0.01	0.52	-0.10
Lateral	$C_{y\beta}$	$C_{l\beta}$	$C_{n\beta}$	C_{lp}	C_{np}	C_{lr}	C_{nr}	$C_{l\dot{\delta}_a}$	$C_{n\dot{\delta}_a}$	$C_{y\dot{\delta}_r}$	$C_{l\dot{\delta}_r}$	$C_{n\dot{\delta}_r}$		
M = 0.257														
Sea level	-1.17	-0.175	0.50	-0.285	-0.14	0.265	-0.75	0.039	0.0042	0.208	0.045	-0.16		
M = 1.8														
55,000 ft	-1.0	-0.09	0.24	-0.27	-0.09	0.15	-0.65	0.017	0.0025	0.05	0.008	-0.04		

Note: All derivatives are per radian.

Figure 5: Aerodynamic data for the aircraft. Any aerodynamic coefficients not shown in the figure can be assumed to be zero.

6.3 Atmospheric Properties



Appendix B
Properties of the U.S. Standard Atmosphere

In this appendix the properties of the 1962 U.S. Standard Atmosphere are tabulated in accordance with Eqs. (1.5–1.17) and (1.21–1.22) of Chapter 1. Two tables are given: Table B1 in English units and Table B2 in Metric units.

Table B1 U.S. Standard atmosphere in English units

Altitude	Temperature	Temperature ratio	Pressure	Pressure ratio	Density	Density ratio	Coefficient of viscosity	Speed of sound
h, ft	$T, ^\circ\text{R}$	θ	P, psf	δ	ρ slugs $\frac{\text{lb}}{\text{ft}^3}$ $\times 10^{-3}$	σ	μ $\frac{\text{lb} \cdot \text{sec}}{\text{ft}^2}$ $\times 10^{-7}$	$a, \text{ft/s}$
0	518.7	1.000	2116	1.000	2.377	1.000	3.737	1,116.4
1,000	515.1	0.9932	2041	0.9644	2.3081	0.97106	3.717	1,112.6
2,000	511.5	0.9863	1963	0.9298	2.2409	0.94277	3.697	1,108.7
3,000	508.0	0.9794	1879	0.8962	2.1751	0.91512	3.677	1,104.0
4,000	504.4	0.9725	1828	0.8637	2.1109	0.88809	3.657	1,101.0
5,000	500.8	0.9657	1761	0.8320	2.0481	0.86167	3.636	1,097.1
6,000	497.3	0.9588	1696	0.8014	1.9868	0.83586	3.616	1,093.2
7,000	493.7	0.9519	1633	0.7716	1.9268	0.81064	3.596	1,089.2
8,000	490.1	0.9459	1572	0.7428	1.8683	0.78602	3.575	1,085.3
9,000	486.6	0.9382	1513	0.7148	1.8111	0.76196	3.555	1,081.4
10,000	483.0	0.9313	1456	0.6877	1.7533	0.73848	3.534	1,077.4
11,000	479.4	0.9244	1400	0.6614	1.7008	0.71555	3.513	1,073.4
12,000	475.9	0.9175	1345	0.6360	1.6476	0.69317	3.493	1,069.4
13,000	472.3	0.9107	1293	0.6113	1.5957	0.67133	3.472	1,065.4
14,000	468.7	0.9038	1243	0.5857	1.5451	0.65003	3.451	1,061.4
15,000	465.2	0.8969	1194	0.5643	1.4956	0.62924	3.430	1,057.3
16,000	461.6	0.8900	1147	0.5420	1.4474	0.60896	3.409	1,053.2
17,000	458.0	0.8831	1101	0.5203	1.4004	0.58919	3.388	1,049.2
18,000	454.5	0.8763	1057	0.4994	1.3546	0.56991	3.366	1,045.1
19,000	450.9	0.8694	1014	0.4791	1.3100	0.55112	3.345	1,041.0
20,000	447.3	0.8625	972	0.4595	1.2664	0.53281	3.324	1,036.8
21,000	443.8	0.8556	932	0.4406	1.2240	0.51497	3.302	1,032.7
22,000	440.2	0.8488	894	0.4223	1.1827	0.49758	3.281	1,028.5
23,000	436.6	0.8419	856	0.4046	1.1425	0.48065	3.259	1,024.4
24,000	433.1	0.8350	820	0.3876	1.1033	0.46417	3.238	1,020.2
25,000	429.5	0.8281	785	0.3711	1.0651	0.44812	3.216	1,016.1
26,000	426.0	0.8213	752	0.3522	1.0280	0.43250	3.194	1,011.7

(continued)

Figure 6: Properties of the US Standard Atmosphere

6.4 Longitudinal Derivatives Formulas

Summary of longitudinal derivatives	
$X_u = \frac{-(C_{D_u} + 2C_{D_0})QS}{mu_0} (s^{-1})$	$X_w = \frac{-(C_{D_w} - C_{L_0})QS}{mu_0} (s^{-1})$
$Z_u = \frac{-(C_{L_u} + 2C_{L_0})QS}{mu_0} (s^{-1})$	
$Z_w = \frac{-(C_{L_w} + C_{D_0})QS}{mu_0} (s^{-1})$	$Z_{\dot{w}} = -C_{z\dot{w}} \frac{c}{2u_0} QS / (u_0 m)$
$Z_{\dot{u}} = u_0 Z_w (ft/s^2) \text{ or } (m/s^2)$	$Z_{\dot{w}} = u_0 Z_w (ft/s) \text{ or } (m/s)$
$Z_q = -C_{zq} \frac{c}{2u_0} QS/m (ft/s) \text{ or } (m/s)$	$Z_{\dot{w}_e} = -C_{z\dot{w}_e} QS/m (ft/s^2)$
$M_u = C_{m\dot{u}} \frac{(QSc)}{u_0 I_y} \left(\frac{1}{ft \cdot s} \right) \text{ or } \left(\frac{1}{m \cdot s} \right)$	
$M_w = C_{m\dot{w}} \frac{(QSc)}{u_0 I_y} \left(\frac{1}{ft \cdot s} \right) \text{ or } \left(\frac{1}{m \cdot s} \right)$	$M_{\dot{w}} = C_{m\dot{w}} \frac{\bar{c}}{2u_0} \frac{QSc}{u_0 I_y} (ft^{-1})$
$M_{\dot{u}} = u_0 M_w (s^{-2})$	$M_{\dot{w}} = u_0 M_w (s^{-1})$
$M_q = C_{mq} \frac{\bar{c}}{2u_0} (QSc)/I_y (s^{-1})$	$M_{\dot{w}_e} = C_{m\dot{w}_e} (QSc)/I_y (s^{-2})$

Figure 7: Summary of longitudinal derivatives

6.5 Longitudinal Flying Qualities

Longitudinal flying qualities				
Phugoid mode				
	Level 1	$\zeta > 0.04$		
	Level 2	$\zeta > 0$		
	Level 3	$T_2 > 55 \text{ s}$		
Short-period mode				
Categories A and C			Category B	
	ζ_{sp}	ζ_{sp}	ζ_{sp}	ζ_{sp}
Level	min	max	min	max
1	0.35	1.30	0.3	2.0
2	0.25	2.00	0.2	2.0
3	0.15	—	0.15	—

Figure 8: Longitudinal Flying Qualities

6.6 MATLAB Code

Here is the Matlab code used for Report and Discussion.

```

1  C_L = 0.735;
2  C_D = 0.263;
3  C_L_u = 0;
4  C_D_u = 0;
5  C_m_u = 0;
6  C_L_0 = C_L;
7  C_D_0 = C_D;
8  C_L_alpha = 3.44;
9  C_D_alpha = 0.45;
10 C_m_alpha = -0.64;
11 C_L_alpha_dot = 0.0;
12 C_m_alpha_dot = -1.6;
13 C_L_q = 0.0;
14 C_m_q = -5.8;
15 C_L_M = 0.0;
16 C_D_M = 0.0;
17 C_m_M = 0.0;
18 C_L_delta = 0.68;
19 C_m_delta = -1.46;
20 %All derivatives are 1/radian
21
22 g = 32.17; %ft/s^2
23 W = 16300; %lb
24 m = W/g; %slug
25 I_x = 3549; %slug*ft^2
26 I_y = 58611; %slug*ft^2
27 I_z = 59669; %slug*ft^2
28 I_xz = 0;
29
30 S = 196.1; %ft^2
31 S_t = 50; %ft^2
32 b = 21.94; %ft^2
33 c_bar = 9.55; %ft
34 c = c_bar; %ft
35
36 C_L_alpha_t = 3;
37 tau = 0.5; %elevetor effectiveness
38 eta = 0.95; %dynamic pressure ratio
39
40 Mach = 0.257;
41
42 %Since it is at sea level, air density is assumed to be at standard
    condistion
43 rho_0 = 0.002377; %slug/ft^3
44 %So is air speed
45 a_0 = 1116.4; %ft/s
46 v_inf = a_0*Mach; %ft/s
47 %We assume that angle of attack, alpha, is zero
48 u_0 = v_inf;
49
50 Q = 0.5*rho_0*u_0^2;
51
52 X_u = -(C_D_u + 2*C_D_0)*Q*S / (m*u_0);

```

```

53 X_w = -(C_D_alpha - C_L_0)*Q*S / (m*u_0);
54 X_delta = 0;
55 Z_u = -(C_L_u + 2*C_L_0)*Q*S / (m*u_0);
56 Z_w = -(C_L_alpha + C_D_0)*Q*S / (m*u_0);
57 C_Z_delta = -C_L_alpha_t*tau*eta*(S_t/S);
58 Z_delta = -C_Z_delta*Q*S / m;
59 M_u = C_m_u * Q*S*c / (u_0*I_y);
60 M_w = C_m_alpha*Q*S*c_bar / (u_0*I_y);
61 M_q = C_m_q*c_bar^2*Q*S / (2*u_0*I_y);
62 M_w_dot = C_m_alpha_dot*c_bar^2*Q*S / (2*u_0^2*I_y);
63 M_delta = C_m_delta*Q*S*c_bar / I_y;
64
65 %3.1
66 A = [X_u, X_w, 0, -g;
67      Z_u, Z_w, u_0, 0;
68      M_u + M_w_dot*Z_u, M_w + M_w_dot*Z_w, M_q + M_w_dot*u_0, 0;
69      0, 0, 1, 0];
70
71
72 B = [X_delta;
73      Z_delta;
74      M_delta + M_w_dot*Z_delta;
75      0];
76
77 %3.2
78 eig_values = eig(A);
79
80 figure(3);
81 plot(real(eig_values), imag(eig_values), 'o', 'MarkerSize', 10, 'LineWidth
82      ', 2);
83 grid on;
84 xlabel('Re');
85 ylabel('Im');
86 title('Eigenvalue Plot');
87 axis equal;
88
89 %Phugoid
90 Phugoid = eig_values(3);
91
92 w_np = sqrt(real(Phugoid)^2 + imag(Phugoid)^2);
93 zeta_p = - real(Phugoid)/w_np;
94 w_dp = imag(Phugoid);
95 tau_p = -1/real(Phugoid);
96
97 %Short
98 Short = eig_values(1);
99
100 w_ns = sqrt(real(Short)^2 + imag(Short)^2);
101 zeta_s = - real(Short)/w_ns;
102 w_ds = imag(Short);
103 tau_s = -1/real(Short);
104
105 %3.3
106 C = eye(4);

```



```

106 D = zeros(4,1);
107 t = 0:0.1:200;
108 u = zeros(1, length(t));
109 x_0 = [0; 0; 0; 0.1];
110 sys = ss(A,B,C,D);
111 [y,t,x] = lsim(sys,u,t,x_0);
112
113 %3.3_1
114 figure(1)
115 subplot 411
116 plot(t,x(:,1));
117 xlabel('time (sec)')
118 ylabel('$\Delta u$ (ft/s)', 'Interpreter','latex')
119 subplot 412
120 plot(t,x(:,2));
121 xlabel('time (sec)')
122 ylabel('$\Delta w$ (ft/s)', 'Interpreter','latex')
123 subplot 413
124 plot(t,x(:,3));
125 xlabel('time (sec)')
126 ylabel('$\Delta q$ (ft/s)', 'Interpreter','latex')
127 subplot 414
128 plot(t,x(:,4));
129 xlabel('time (sec)')
130 ylabel('$\Delta \dot{\theta}$ (rad/s)', 'Interpreter','latex')
131
132 x_0 = [0;0;0;0];
133 u = ones(1, length(t));
134 [y,t,x] = lsim(sys,u,t,x_0);
135 %3.3_2
136 figure(2)
137 subplot 411
138 plot(t,x(:,1));
139 xlabel('time (sec)')
140 ylabel('$\Delta u$ (ft/s)', 'Interpreter','latex')
141 subplot 412
142 plot(t,x(:,2));
143 xlabel('time (sec)')
144 ylabel('$\Delta w$ (ft/s)', 'Interpreter','latex')
145 subplot 413
146 plot(t,x(:,3));
147 xlabel('time (sec)')
148 ylabel('$\Delta q$ (ft/s)', 'Interpreter','latex')
149 subplot 414
150 plot(t,x(:,4));
151 xlabel('time (sec)')
152 ylabel('$\Delta \dot{\theta}$ (rad/s)', 'Interpreter','latex')
153
154 %3.5
155 A_sh = [Z_w, u_0; M_w + M_w_dot*Z_w, M_q + M_w_dot*u_0];
156 A_ph = [X_u, -g; -Z_u/u_0, 0];
157
158 eig_values_ph = eig(A_ph);
159 eig_values_sh = eig(A_sh);

```

```
160
161 %Approximated Phugoid
162 Phugoid_a = eig_values_ph(1);
163
164 w_np_a = sqrt(real(Phugoid_a)^2 + imag(Phugoid_a)^2);
165 zeta_p_a = - real(Phugoid_a)/w_np;
166 w_dp_a = imag(Phugoid_a);
167 tau_p_a = -1/real(Phugoid_a);
168
169 %Approximated Short
170 Short_a = eig_values_sh(1);
171
172 w_ns_a = sqrt(real(Short_a)^2 + imag(Short_a)^2);
173 zeta_s_a = - real(Short_a)/w_ns;
174 w_ds_a = imag(Short_a);
175 tau_s_a = -1/real(Short_a);
```

Listing 1: Matlab Code used for Result and Discussion